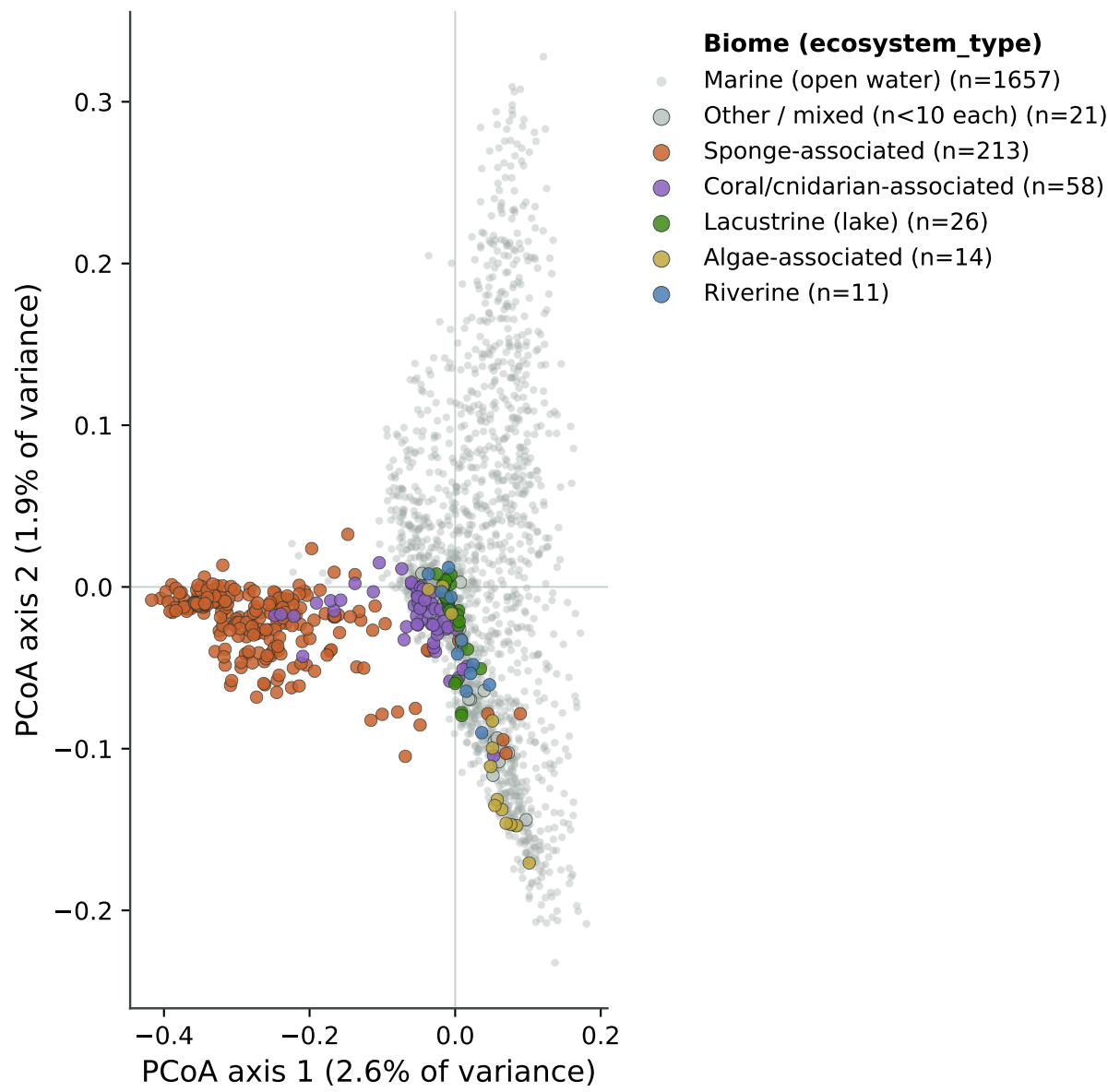


Site ordination by PhaC gene-cluster composition (Jaccard dissimilarity, PCoA)



n = 2000 sites (richest 2,000 of qualifying sites, min. 3 distinct clusters/site) | total variance explained (2 axes) = 4.5%
Mantel test (Jaccard dissimilarity vs. geographic distance): $r = 0.102$, $p = 0.001$ (999 permutations) — significant but weak isolation-by-distance signal